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Paleomagnetic results from a reconnaissance study of Santiago (Cape Verde Islands): Identification of cryptochron C2r.2r-1

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ABSTRACT

New paleomagnetic data from three lava sequences on Santiago, Cape Verde Islands, are presented. The paleomagnetic data are coupled with $^{40}Ar/^{39}Ar$-age determinations, allowing detailed correlations between the three profiles and the Geomagnetic Polarity Time Scale (GPTS). The younger of the profiles, the Ribeira da Barca profile, recorded flows of normal polarity that correlate with the Brunhes Chron, whereas the Porto Formosa profile of reverse polarity correlates with the lower Matuyama Chron. The São Gonçalo profile consists of both reverse- and normal-polarity flows that belong to the Matuyama and Gauss Chron, respectively. Virtual geomagnetic poles (VGPs) from a sequence of flow units in the São Gonçalo profile classify as transitional, and we interpret it as a geomagnetic event (SG-I). Well-defined $^{40}Ar/^{39}Ar$ ages bound the occurrence of event SG-I to the interval 2.37–2.43 Ma ($\pm 2\sigma$), which is synchronous with the timing of cryptochron C2r.2r-1 in the marine magnetic anomaly record. This study therefore confirms the existence of cryptochron C2r.2r-1 and indicates that it was a global-scale phenomenon. The VGPs indicate that the field was in a meta-stable transitional state during cryptochron C2r.2r-1, and that it spent some time in or around the paleomagnetic transitional VGP-cluster patch close to Australia.

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1. Introduction

The importance of a widespread geographical coverage of paleomagnetic observations, preferably from volcanic rocks (e.g. Camps et al., 2001), when studying general features of the geomagnetic field have encouraged many paleomagnetists to study remote areas of Earth devoid of paleomagnetic data, including isolated oceanic islands dispersed around the globe where volcanic rocks are highly abundant. The abundance of Pleistocene and Pliocene volcanic rocks makes the Cape Verde archipelago particularly interesting for the ongoing study of the 0–5 Ma geomagnetic field, focusing on paleosecular variation (PSV) and the time-averaged field (TAF) (Johnson et al., 1998; Johnson and Constable, 1995). Such lava sequences are also essential for the study of the existence and origin of geomagnetic excursions and other short-term instabilities of the magnetic field, such as cryptochrons. Paleomagnetic records of excursions and cryptochrons from lava flows and sedimentary sections are not only of considerable importance for geological and stratigraphic studies, but they also provide important insights into the geodynamo processes generating the Earth's magnetic field (e.g. Gubbins, 1999; Glatzmaier et al., 1999). Currently, a large effort is being made to date the lava flows that recorded these short geomagnetic instabilities during the Brunhes and Matuyama Chrons (Singer et al., 1999, 2002; Coe et al., 2004; Tauxe et al., 2004).

The term cryptochron was introduced by Cande and Kent (1992) to describe tiny wiggles in magnetic anomaly records from the world's major ocean-ridge systems that represent either very short polarity intervals, excursions, or intensity fluctuations of the dipole field (Cande and Kent, 1992, 1995). Cryptochrons were not included in the Geomagnetic Polarity Time Scale (GPTS), since Cande and Kent (1992, 1995) used a 30 kyr cut-off to separate well-characterized anomalies (subchrons), which most probably reflect true geomagnetic polarity reversals, from small-scale anomalies of uncertain origin. Another reason for not including tiny wiggles in the GPTS was that they were not well documented in magnetostratigraphic sections (Cande and Kent, 1992, 1995). To confirm the existence and understand the origin of cryptochrons, it is therefore important to identify them in terrestrial magnetostratigraphic records.

In this study, we present new paleomagnetic results from three lava sequences on Santiago, which is the largest of the islands in the Cape Verde archipelago. The three profiles are temporally constrained by $^{40}Ar/^{39}Ar$ ages (Holm et al., 2008), which allow

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correlation to the GPTS and discussion of their possible implications for the geomagnetic field structure.

2. Geological setting and paleomagnetic sampling

The island of Santiago belongs to the Cape Verde archipelago, located in the Atlantic ocean some 500–800 km off the coast of Senegal. Hotspot activity around 19–20 Ma (McNutt, 1988; Klerkx and De Paepe, 1976), or, alternatively, around 21–22 Ma (Holm et al., 2008), resulted in a large crustal feature rising 2 km above the oceanic floor, the so-called Cape Verde Rise, on which the Cape Verde Islands are situated (Gerlach et al., 1988). The age of the underlying crust ranges from 122 to 140 Ma, corresponding to Mesozoic marine anomalies M1–M16 (Hayes and Rabinowitz, 1975; Stillman et al., 1982). The very slow eastward movement of the African lithospheric plate (Pollitz, 1991) relative to the mantle plume cannot explain the two linear chains, of altogether 10 major and some minor islands, that constitute the Cape Verde archipelago (Fig. 1a). The formation of the islands is believed to be related to either the geometry of the flow of plume material, as envisaged by Holm et al. (2006), or to the structure of the lithosphere (Williams et al., 1990). The age of the igneous exposures on the islands range from tilted and uplifted pillow lavas of MORB-character overlain by Jurassic sediments (exposed only on Maio) (Stillman et al., 1982) to the historical eruptions on Fogo, where the most recent eruption took place in 2000 AD. Generally, ankaramite and silica-undersaturated lavas, such as nephelinite and basanite, characterize the hotspot volcanics encountered on all the islands (Gerlach et al., 1988).

The paleomagnetic data presented in this paper stem from the largest island in the Cape Verde archipelago, the island of Santiago (Fig. 1b), which covers an area of 992 km^2. The island is irregular in shape and dominated by well-exposed lava flows and a high topographical relief. It is highly dissected by deep valleys indicating periods of low volcanic activity (Holm et al., 2008). According to Serralheiro (1976), the oldest exposures on Santiago are located in the central part of the island (Fig. 1b), mainly consisting of lava flows from the late Cretaceous or early Tertiary (Old Volcanic Complex). Serralheiro (1976) found that the bulk of the island comprises late Miocene (Flamengos Formation) and Pliocene (Pico da Antonio Formation) lava flows, the latter occasionally found intercalated with calcarenites. The youngest volcanic exposures on Santiago are the well-preserved scoria cones (Monte das Vacas Formation), which may be observed in several areas of the island (Serralheiro, 1976). However, the ages assigned to these formations (Serralheiro, 1976), based on stratigraphical and paleontological evidence, have been superseded by a modern $^{40}Ar/^{39}Ar$ study by Holm et al. (2008), which generally yield ages younger than those suggested by Serralheiro (1976). These new results indicate that the dominating volcanic exposures in the southern and northern parts of the island both have ages around 2–3 Ma (Holm et al., 2008).

Only a few paleomagnetic studies have previously been undertaken on the Cape Verde archipelago (Watkins et al., 1968; Storetvedt and Løvlie, 1983; Abranches et al., 1990; Knudsen and Abrahamsen, 2000; Knudsen et al., 2003, 2005) and so far only Watkins et al. (1968) and Abranches et al. (1990) have undertaken such studies on Santiago. Abranches et al. (1990) mainly collected samples from the Old Volcanic Complex and the Flamengos lavas, with the focus being on magnetomineralogy. Watkins et al. (1968) conducted paleomagnetic studies on seven of the Cape Verde Islands, including Santiago, and subsequently concluded that the central part of Santiago is dominated by flows of normal polarity, while the flows in the southern part dominantly are of reverse polarity. The only overlap between samples collected in this study and earlier studies are those collected by Watkins et al. (1968) in the Ribeira da Barca valley.

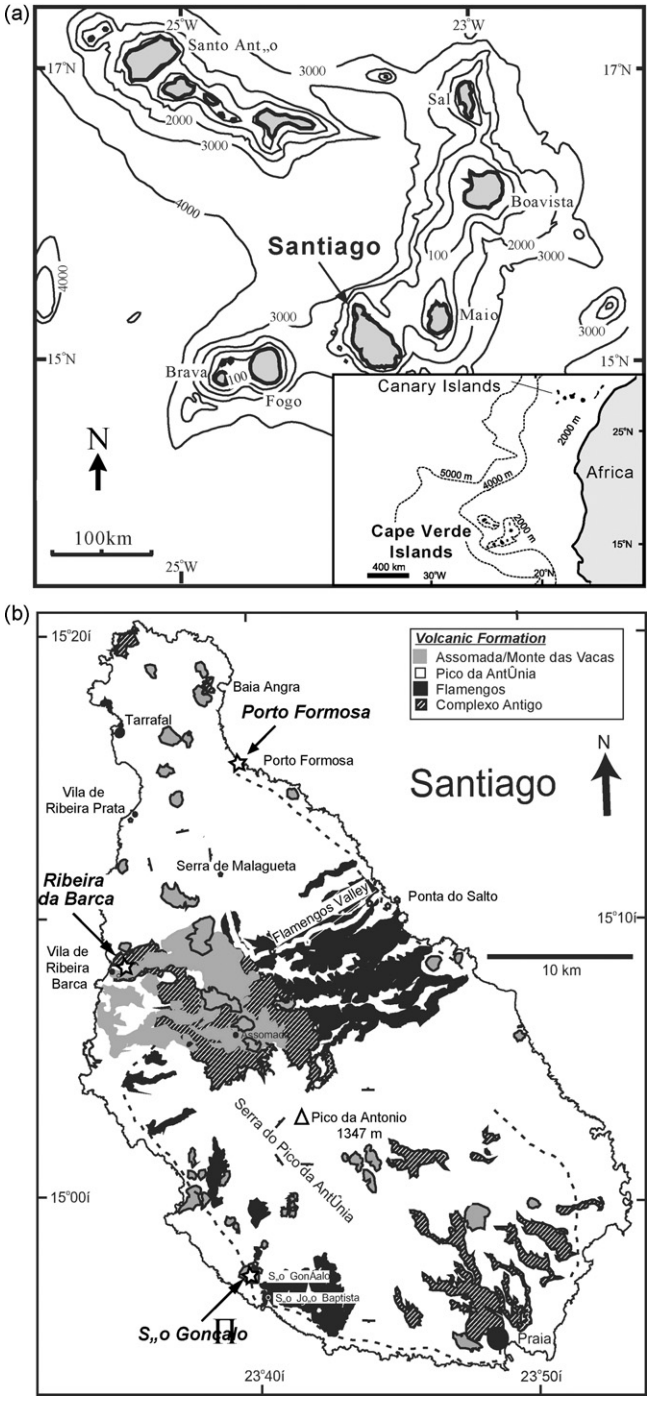


Fig. 1. (a) Bathymetric map of the Cape Verde archipelago. The inserted map shows the location of the archipelago relative to West Africa. (b) The island of Santiago with the geographical locations of the São Gonçalo profile (Lat. 14°57′45″N, Long. 23°40′01″W), the Ribeira da Barca profile (Lat. 15°08′09″ N, Long. 23°44′35″ W), and the Porto Formosa profile (Lat. 15°15′26″ N, Long. 23°40′51″W). The map shows the volcanic formations on the island.

2.1. Sampling

We have studied three lava sequences on Santiago: a ∼ 150 m thick sequence of flat-lying lavas at São Gonçalo near the village of São João Baptista, a 130 m sequence of lavas above Ribeira da Barca on the west coast, and a 100 m thick sequence of mainly marine volcanics at Porto Formosa on the northern part of the east coast (Fig. 1b).

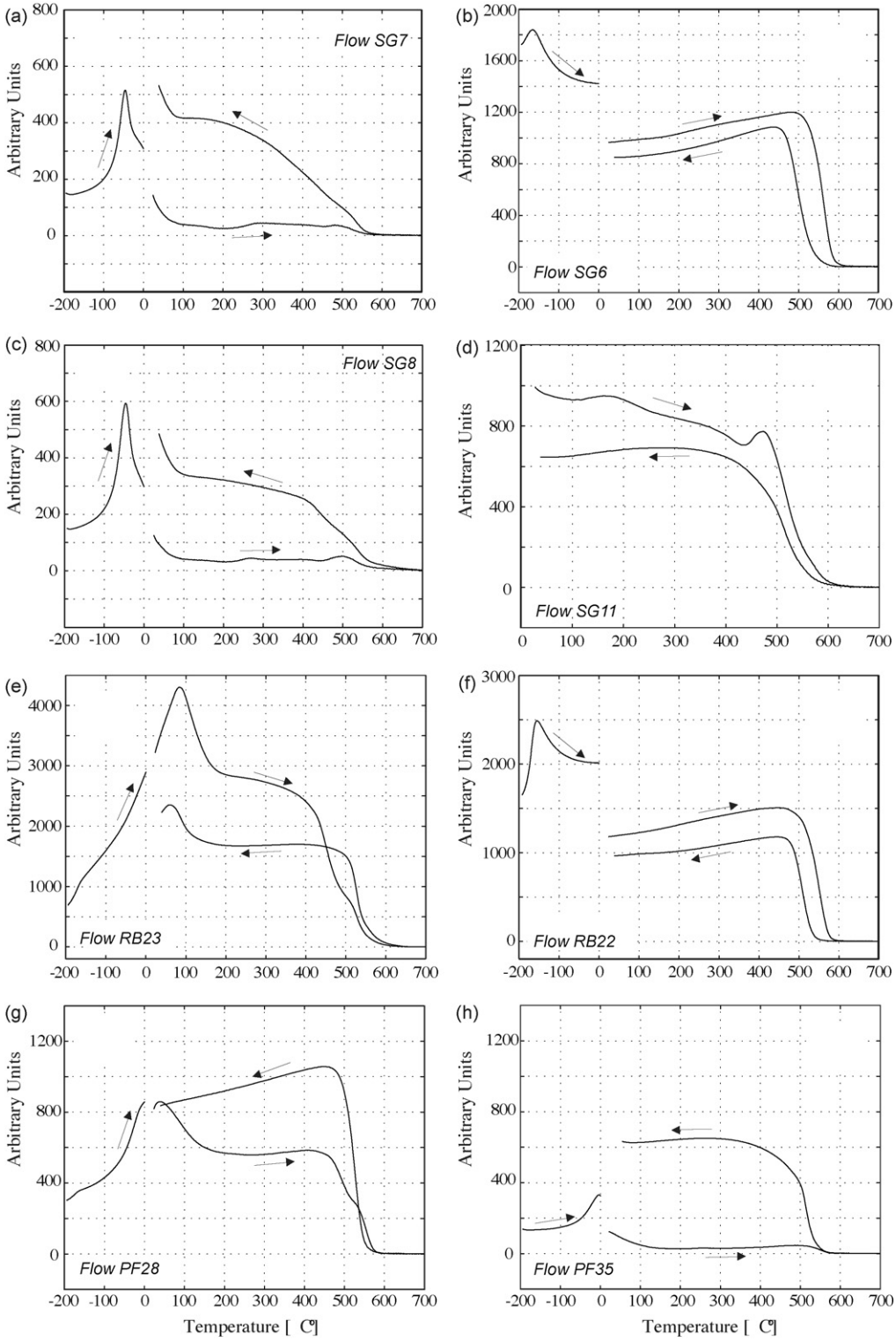


Fig. 2. Weak-field magnetic susceptibility as a function of temperature. The thermomagnetic behavior of the São Gonçalo flows range from almost single-phase titanium-rich titanomagnetite (~TM60-TM70) with an irreversible heating-curve after high-temperature oxidation (a) and (c) to single-phase titanium-poor titanomagnetite with a reversible heating-curve and presence of a Verwey transition (b). The behavior of flow SG11 is diagnostic of a titatium-poor titanomagnetite (d). The Ribeira da Barca flows show an almost similar behavior with some containing both titanium-rich and a titanium-poor titanomagnetite (e) and others only titanium-poor titanomagnetite (f). Except for the subaerial flow presented in (g), which has both a titanium-rich and titanium-poor titanomagnetite phase, the Porto Formosa profile consists of hyaloclastites characterized by low susceptibility and single-phase titanium-rich titanomagnetite as the dominant magnetic mineral (h).

The São Gonçalo lava sequence constitutes the entire local sequence of lava flows belonging to the Pico da Antonia Formation. The lava profile is overlain by two monogenetic scoria cones of the Monte das Vacas Formation (Alves et al., 1979). The paleomagnetic profile consists of two sections, containing a total of 16 aphanitic lava flows with a highly varying thickness (2–30 m). A steep slope of flat-lying, well-exposed lavas constitute the uppermost section, whereas the 3 flows in the bottom section dip 2–5 ° towards the

Table 1
A summary of the $^{40}Ar/^{39}Ar$ ages from the paleomagnetic profiles. Details are available as auxiliary material and are presented in full in Holm et al. (2008). The plateau age, number of steps and initial $^{40}Ar/^{36}Ar$ ratio for the $^{40}Ar/^{39}Ar$ ages derived from flow units included in the paleomagnetic profiles. The uncertainties presented are $\pm 2\sigma$ and GM indicates ground mass phase. The data were calibrated using the FCT-3 standard of Renne et al. (1994).

Profile	Flow	Phase	Plateau age (Ma) $\pm 2\sigma$	N	$\%^{39}Ar$
São Gonçalo	SG18	GM	2.25 ± 0.09	6	73.3
São Gonçalo	SG11	GM	2.41 ± 0.04	5	79.4
São Gonçalo	SG7	GM	2.38 ± 0.05	5	87.6
São Gonçalo	SG6	GM	3.25 ± 0.04	5	64.7
Ribeira da Barca	RB21	GM	0.74 ± 0.04	5	66.7
Porto Formosa	PF28	GM	2.40 ± 0.02	4	64.8
Porto Formosa	PF35	GM	2.50 ± 0.02	4	79.2

south, on what appears to be an original slope. The uppermost flow unit in the top section, SG19, may be separated from the underlying flows by an unconformity, but field observations do not indicate a significant time gap between SG19 and the underlying flows. The two sections are separated by an unconformity, but appear to be in stratigraphic succession without any overlap. This observation is supported by $4^{40}Ar/^{39}Ar$ ages that were used to constrain the stratigraphy of this profile.

The profile at Ribeira da Barca consists of three lava-flow sections, of which only the bottom two were sampled in this paleomagnetic study. Flows collected in the lowermost section have a considerable thickness (~20–30 m), while the middle section consists of thin (~2–3 m), weathered lava flows. All 9 flow units in this profile are nearly horizontal, aphanitic, and mostly aphyric, lavas.

At Porto Formosa, samples were collected from 8 flow units (Fig. 1b). Except for the flat-lying top flow, which is subaerial and 5 m thick, the profile consists of pillow basalts and hyaloclastites with an average thickness of 0.5–1 m (Fig. 2). All these flow units belong to the Pico da Antonio Formation (Holm et al., 2008). The pillow basalts and hyaloclastites dip at ~ 25° towards the east, on what appears to be an original slope.

Cores were collected from 33 flow units, with an average of ~ 3 cores from each flow. All cores were oriented by magnetic compass, and the majority (~70%) also by sun compass. In general, orientations obtained by magnetic compass and sun compass differed by less than ~ 3°. Since none of the flows appear to have been subjected to tectonic disturbance after acquisition of the magnetic remanence, no tilt correction has been applied.

3. $^{40}Ar/^{39}Ar$ ages

The three paleomagnetic profiles are constrained in time by seven $^{40}Ar/^{39}Ar$ ages, which are based on samples that were collected together with the paleomagnetic samples. The samples were jaw-crushed and sieved to a fraction of 75–200 or 200–300 μm, before removal of the phenocrysts by magnetic and heavy-liquid separation, and final hand picking under optical microscope. The ground-mass material from the samples was subsequently analyzed by the incremental step-heating $^{40}Ar/^{39}Ar$ method. FCT-3 biotite standards were used for calibration, assuming an age of 28.04 ± 0.01 Ma (Renne et al., 1994). The incremental heating experiments were performed through 6–13 temperature steps from 500 ° C to 1400 °C. An age from each temperature step was calculated, assuming an initial atmospheric $^{40}Ar/^{36}Ar$ ratio of 295.5. Plateau ages were accepted for all seven samples, with each plateau consisting of at least three adjacent temperature steps whose ages overlap within the uncertainty at the 95% confidence level and

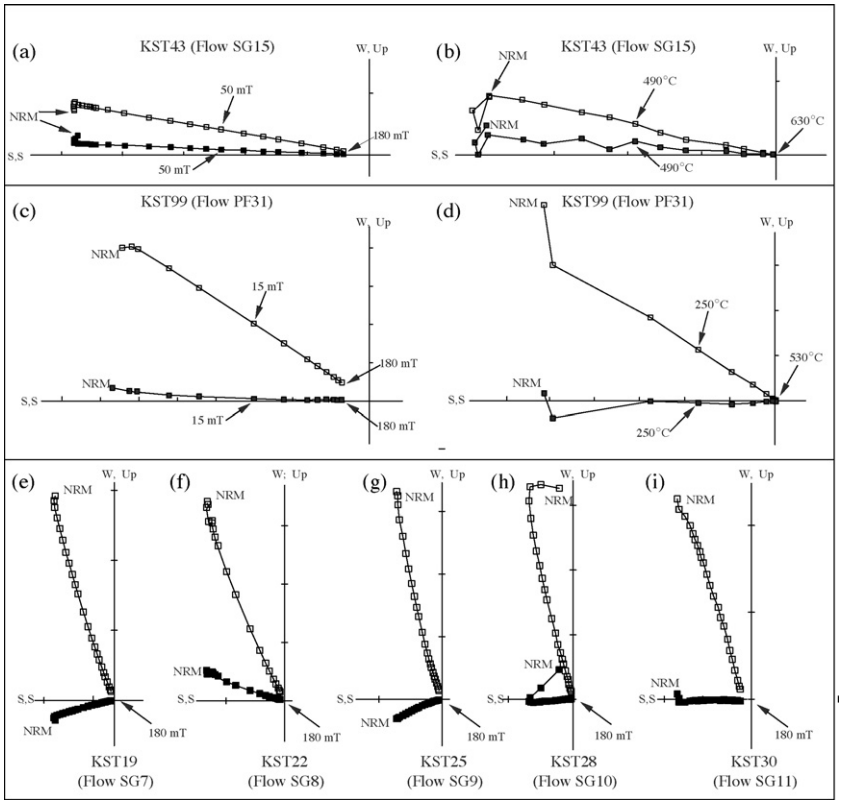


Fig. 3. Zijderveld diagrams of sample KST43 (flow SG15) after (a) alternating field (AF) and (b) thermal demagnetization, respectively. Zijderveld diagrams of sample KST99 (flow PF31) after (c) AF demagnetization and (d) thermal demagnetization. The ChRM directions were generally easily defined and the directions obtained after AF and thermal demagnetization proved consistent. Overall, the AF demagnetization was more effective in producing stable NRM decays towards the origin of the Zijderveld diagrams. Diagrams e-i show the demagnetization behavior of the five flows that define event SG-I.

represent at least 50% of the ^{39}Ar released during the step-heating process. The plateau ages (listed in Table 1) were in all seven cases preferred over the associated isochron ages (Holm et al., 2008). A more detailed description of the ^{40}Ar/^{39}Ar study is presented in Holm et al. (2008), whereas age spectra and isochron plots for all seven samples are available as auxiliary material.

4. Experimental procedures and results

4.1. Magnetic mineralogy

Samples from all flows were subjected to measurements of the magnetic susceptibility as a function of temperature in order to reveal the Curie temperature (T_C) diagnostic of the magnetic minerals in the flows. The flows in the São Gonçalo profile display behavior consistent with a varying titanomagnetite composition. Compositions range from single-phase titanium-rich titanomagnetite (∼TM60-TM70) in a low oxidation state (flows SG7, SG8, SG9, SG13, and SG16), with no indication of a Verwey transition (Fig. 2a and c), to single-phase titanium-poor titanomagnetite (∼TM0-TM5) with a $T_C$of ∼ 550° and clear identification of a Verwey transition (Fig. 2b). In between these two compositions we find flows with two phases of titanomagnetite, i.e. a titanium-poor and titanium-rich phase. The thermomagnetic behavior of flow SG11 indicates that titanium-poor titanomagnetite is the dominant magnetic mineral in this flow (Fig. 2d). We interpret the irreversibility of the curves in (a) and (c) as indicating exsolution of the titanium-rich phase upon heating. The thermomagnetic curves for the Ribeira da Barca lava sequence indicate compositions similar to those observed for the São Gonçalo flows. None of Ribeira da Barca flows, however, display single-phase titanium-rich titanomagnetite as the dominant magnetic mineral, but rather compositions varying from two-phase titanomagnetite (Fig. 2e), characterized by two diagnostic Curie temperatures of ∼ 100 ° C and ∼ 480 °C, to single-phase titanium-poor titanomagnetite (Fig. 2f).

The Porto Formosa flows differ somewhat in composition from the flows in the two other profiles. This is probably due to the presence of a glass phase in the chilled submarine rocks. The subaerial flow in the very top of the profile is dominated by both a titanium-rich and a titanium-poor titanomagnetite phase (Fig. 2g). The irreversible thermomagnetic behavior, typical of the hyaloclastites (Fig. 2h), reflects a single-phase titanium-rich titanomagnetite with ∼ TM80 as the dominant magnetic mineral. Such a composition can only be preserved as single-phase if cooled rapidly, typically by extrusion into sea water, which is in accordance with the origin of the hyaloclastites in the profile.

4.2. Demagnetization of the NRM

The direction of the characteristic remanent magnetization (ChRM) was determined by principal component analysis (PCA) (Kirschvink, 1980) after step-wise alternating field (AF) demagnetization (0–180 mT using 24, 17, or 13 steps) of all samples from the three lava sequences. Several companion samples were additionally subjected to step-wise thermal demagnetization (with intervals varying from T_{room} − 530 ° C to T_{room} − 630 °C) for proof of consistency.

The AF demagnetization revealed only little or no evidence of secondary magnetizations, and the ChRMs were in general easily defined. Magnetic overprints were in general easily re moved by thermal demagnetization at temperatures around 100–200 °C. The Zijderveld diagrams in Fig. 3a and c show the behavior of two representative samples subjected to AF demagnetization. Comparison of Fig. 3a,b and Fig. 3c,d, respectively, reflect the general consistency in directions between ChRMs derived from AF demagnetization and thermal demagnetization of specimens from the same core. Five flow units (SG7-SG11) in the São Gonçalo profile display anomalous directions characterized by steep inclinations (Fig. 3e–i). The within-flow scatter for SG7-SG9 is large compared to SG10 and SG11 (Table 2), which agrees with the observed variations in the magnetic mineralogy: titanium-rich titanomagnetite is predominant in flows SG7-SG9, whereas SG10 and SG11 are dominated by titanium-poor titanomagnetite. Note that the altogether 15 directions that derive from these five consecutive flow units (SG7-SG11) seem to cluster around a common mean direction (Fig. 4). The mean declination and inclination of all flow units from the three profiles are listed in Table 2 and shown in Fig. 5, along with the GAD inclination for this area (∼ 28°).

Table 2
Site-mean paleomagnetic directions. Mean declination (Dec) and inclination (Inc) along with the number of samples measured from each flow (N), the corresponding α_{95}, the precision parameter κ, and the coordinates of the mean VGP positions.

São Gonçalo	Dec (°)	Inc (°)	*N*	α_{95}	κ	Plat (°)	Plong (°)
SG19	181.3	−20.2	3	3.4	1295.2	−85.1	320.7
SG18	190.0	−28.5	2	–	1459.1	−80.3	243.6
SG17	199.5	−29.7	2	–	2764.6	−71.2	240
SG16	183.7	−10.5	3	8.5	213.5	−79.6	315.2
SG15	182.0	−5.6	3	6.3	379.2	−77.7	326.6
SG14	186.7	−16.5	3	8.3	223.0	−80.8	289.9
SG13	174.9	−20.2	3	17.4	51.3	−83.2	24.0
SG12	179.1	−30.3	3	7.8	248.0	−88.6	117.9
SG11	182.6	−70.7	3	3.2	1527.1	−50.0	158.3
SG10	168.4	−75.0	3	3.3	1384.0	−42.5	148.6
SG9	176.1	−70.5	3	9.5	169.2	−50.2	152.5
SG8	180.3	−71.0	3	13.2	88.2	−49.6	156.3
SG7	165.7	−78.7	3	14.2	76.9	−36.0	149.5
SG4	6.1	31.2	2	–	108.1	83.9	48.3
SG5	13.5	26.0	2	–	650.4	76.9	69.9
SG6	10.0	27.8	3	4.3	836.0	80.3	66.4
Ribeira da Barca	**Dec (°)**	**Inc (°)**	***N***	**α_{95}**	**κ**	**Plat (°)**	**Plong (°)**
RB27	8.4	7.4	2	–	73.8	76.0	118.9
RB26	27.9	1.6	3	25.6	24.2	59.0	90.7
RB25	15.2	−2.7	3	11.6	114.6	67.8	112.1
RB24	11.3	1.3	2	–	602.8	71.8	117.1
RB23	9.8	1.5	3	3.9	1012.4	72.8	120.9
RB22	6.0	24.7	3	12.3	102.2	83.8	84.6
RB21	2.5	16.0	3	28.4	19.8	82.7	136.0
RB20	21.9	31.9	3	23.0	29.8	68.9	56.8
RB19	354.1	−35.1	2	–	42.5	55.2	165.8
Porto Formosa	**Dec (°)**	**Inc (°)**	***N***	**α_{95}**	**κ**	**Plat (°)**	**Plong (°)**
PF28	203.6	−21.5	4	6.2	220.6	−66.7	252.7
PF29	195.8	−28.9	3	10.0	154.3	−74.8	242.3
PF30	191.2	−29.8	2	–	72.2	−79.3	239.3
PF31	180.8	−31.0	3	4.6	723.7	−88.1	180.0
PF32	186.2	−21.5	3	10.9	128.6	−82.8	177.8
PF33	186.0	−27.1	1	–	–	−84.2	251.6
PF34	184.2	−31.7	2	–	176.0	−85.4	217.3
PF35	175.7	−31.1	3	22.3	31.7	−85.5	89.9

4.3. Paleomagnetic profiles

The three lava sequences are temporally constrained by ^{40}Ar/^{39}Ar ages (Table 1). Fig. 6 shows the average direction and polarity of all flow units at Ribeira da Barca (Fig. 6a), São Gonçalo (Fig. 6b), and Porto Formosa (Fig. 6c), along with the stratigraphic positions of the ^{40}Ar/^{39}Ar ages in each profile. The mean directions of all flow units in each profile are listed in Table 3 along with the common statistical parameters and the associated mean VGP positions.

The lowermost part of the Ribeira da Barca profile is constrained to the Brunhes Chron by one ^{40}Ar/^{39}Ar age, which is in agreement

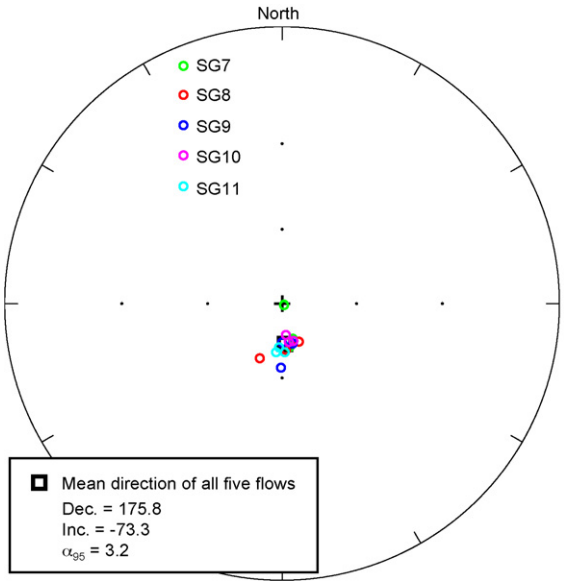


Fig. 4. Stereogram showing the 15 ChRM directions from flows SG7-SG11 in the São Gonçalo profile, which display transitional polarity.

with all flows displaying normal polarity (Fig. 6a). The high angular standard deviation (from VGPs) of 14.4°(+7.4/ − 3.7) is partly due to the somewhat anomalous direction recorded by the lowermost flow in the profile (RB19). If flow RB19 is disregarded, the ASD value drops to 10.8 °.

The uppermost section of the São Gonçalo profile is unequivocally constrained to the lower Matuyama by three $^{40}Ar/^{39}Ar$ ages (Fig. 6a). This is consistent with all flows in this section displaying either reverse or transitional polarity. The lowermost section of three normal-polarity flow units is constrained to the Gauss Chron by one $^{40}Ar/^{39}Ar$ age (Fig. 6a). A reversal test (McFadden and McElhinny, 1990), comparing the mean direction of all normal-polarity flows to the mean direction of all reverse-polarity flows in the profile, is positive with classification C (angle between the two means = 9.4°). This implies that the antipode of the mean reverse-polarity direction is indistinguishable from the mean of the normal-polarity flows at the 95% confidence level. The sequence of 5 consecutive flow units characterized by anomalously steep inclinations, yield VGP latitudes between 35° and 50° (Fig. 7). Adopting the widely used definition of transitional polarity as when the virtual geomagnetic poles (VGP) deviate more than 45° from the geographic pole (Verosub, 1977), we define this sequence of flows as event SG-I. If the flows included in SG-I are disregarded, the VGP angular standard deviation (ASD) of the lava sequence is 8.3°(+3.2/ − 1.8).

The whole span of the Porto Formosa profile is constrained to the lower Matuyama Chron by $^{40}Ar/^{39}Ar$ ages of the upper-

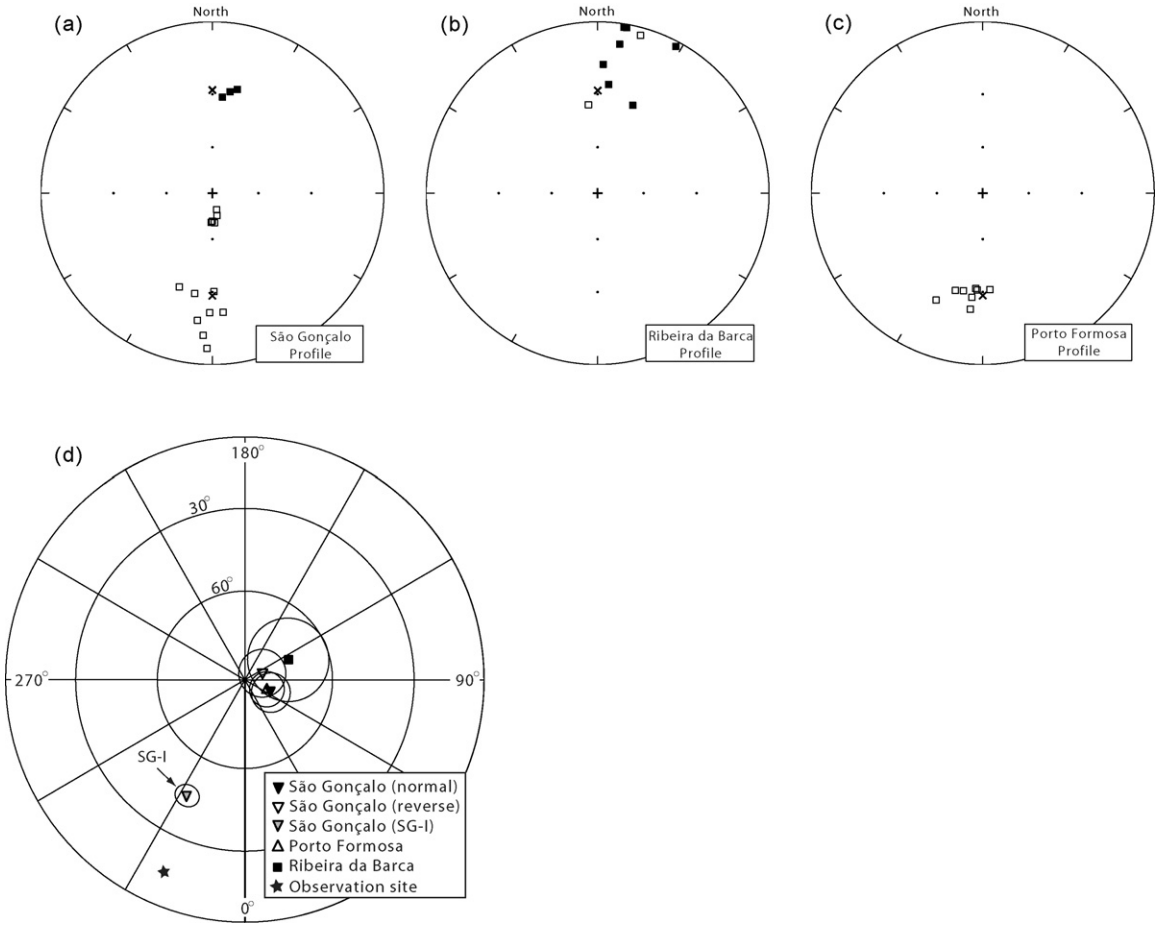


Fig. 5. Stereograms showing the mean directions of all flows in (a) the São Gonçalo profile, (b) the Ribeira da Barca profile, and (c) the Porto Formosa profile. The cross denotes the direction of the GAD field at this location. Paleomagnetic poles based on flow units in the São Gonçalo (SG), Ribeira da Barca (RB), and Porto Formosa (PF) profiles are shown in (d). For the São Gonçalo profile, the paleomagnetic poles are displayed for flows of both normal and reverse polarity. All poles are projected into the northern hemisphere.

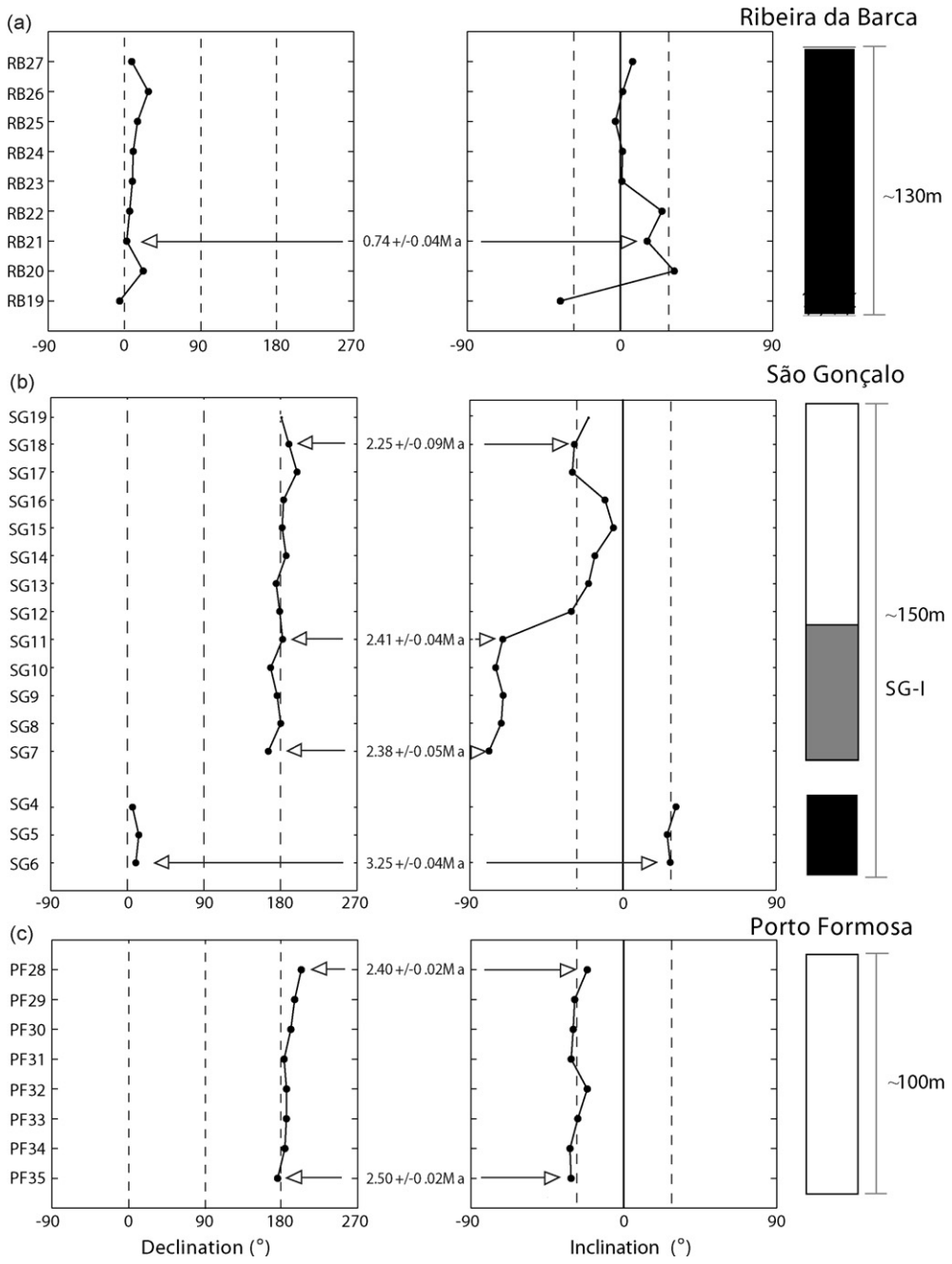


Fig. 6. The mean declination and inclination of the flow units in the stratigraphic sequence for (a) the Ribeira da Barca profile, (b) the São Gonçalo profile, and (c) the Porto Formosa profile. The polarity of the profiles and the labelled magnetic events are shown to the right. Black (white) denotes normal (reverse) polarity, whereas grey denotes transitional polarity. Also shown are the $^{40}Ar/^{39}Ar$ ages (Holm et al., 2008) and the stratigraphic position of the dated flows (age uncertainties are $\pm 2\sigma$). The dashed, vertical lines in the inclination plots denote the inclination (pos. and neg.) expected from a GAD field.

Table 3
Mean VGPs for all profiles and magnetic events. N, number of flows, D_m and I_m, mean declinations and inclinations, and the associated $\alpha_{95}(^\circ)$, κ, and between flow ASD. Plat and Plong, coordinates of mean VGP positions, dp and dm, uncertainties associated with the mean VGP positions. (∗) indicates that all flows with transitional polarity are excluded, whereas (∗∗) indicates that the Brunhes-aged Ribeira da Barca flows additionally are excluded.

Profile	Flows	N	D_m (°)	I_m (°)	α_{95} (°)	κ	ASD (°)	Plat (°)	Plong (°)	dp	dm
São Gonçalo	Reverse	8	184.6	−20.4	8.0	49.5	8.9	83.7	110.4	4.4	8.4
São Gonçalo	Normal	3	10.1	28.4	6.6	351.4	–	80.2	64.0	4.0	7.2
São Gonçalo	SG-I	5	175.8	−73.3	3.2	390.2	6.9	−45.9	153.9	5.1	5.7
Ribeira da Barca	Normal	9	10.8	5.5	13.9	14.6	14.4	73.7	114.5	7.0	13.9
Porto Formosa	Reverse	8	188.4	−28.2	5.9	88.9	8.6	81.9	66.9	3.5	6.5
All*	–	28	8.2	18.9	5.8	23.1	11.7	80.5	98.6	3.1	6.0
All(reverse)**	–	19	6.9	24.9	4.2	64.9	8.4	83.0	81.0	2.4	4.5

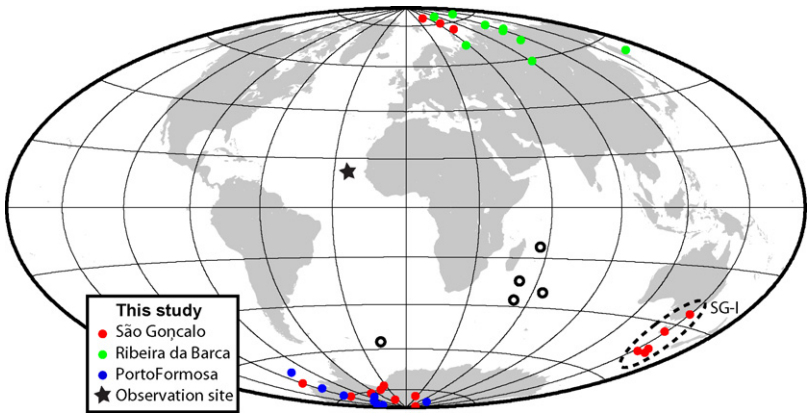


Fig. 7. Virtual geomagnetic poles for all flow units in the São Gonçalo (red), Ribeira da Barca (green), and Porto Formosa (blue) profiles. Also shown is the geographic location of the observation site. The black circles denote VGPs associated with the Halawa excursion observed on Hawaii (Herrero-Bervera et al., 2007). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

most (2.40 ± 0.02 Ma) and lowermost (2.50 ± 0.02 Ma) flow units in the lava sequence (Fig. 6c). All flows in this profile display reverse polarity, and directions cluster around their mean direction (Fig. 5c), which is expressed through the relatively low ASD of 8.6 °(+4.86/ − 2.32). It is noticeable that the seven pillow basalts and hyaloclastites, which dip ∼ 25° towards east, yield a mean inclination (−28.9°) that is very close to the inclination expected for the latitude of this profile (−28.6°). The paleomagnetic data therefore support the interpretation that these flow units were embedded on an original slope and, consequently, that a bedding correction should not be applied. If the dipping flows had originally been embedded as horizontal units, a tectonic correction should have be applied, in which case the mean inclination would be −23.6°, whereas the mean declination would change from 185.7 ° to 198.0°.

5. Discussion

The Ribeira da Barca lava sequence, which correlates with the lower Brunhes Chron, represents the youngest of the profiles sampled on Santiago in this paleomagnetic study. Precise age constraints allow unequivocal correlation of the uppermost section of the São Gonçalo profile and the Porto Formosa profile to the lower Matuyama Chron, and it seems possible that the two profiles overlap (Fig. 8). The $^{40}Ar/^{39}Ar$ age of 3.25 ± 0.04 Ma suggests that the lowermost section of the São Gonçalo profile correlates with a zone of normal polarity, between the Mammoth and Kaena Subchrons, in the Gauss Chron.

It is difficult to determine the degree of overlap between the Porto Formosa profile and the middle section of the São Gonçalo profile. An age of 2.50 ± 0.02 Ma of the lowermost flow in the Porto Formosa profile rules out the possibility that this entire profile is younger than event SG-I. This indicates that either event SG-I occurred in the time span covered by the Porto Formosa profile, but was not recorded because of the stochastic nature of volcanic eruptions, or, perhaps more likely, that the entire Porto Formosa profile is slightly older than event SG-I.

5.1. VGP dispersion and the time-averaged field

The VGP angular standard deviation (ASD) of all non-transitional flows in this paleomagnetic study was determined by reversing the directions of normal polarity and omitting the directions from event SG-I in the São Gonçalo profile. This procedure yielded an ASD value of 11.7° (+2.7/ − 1.9 (±2σ)) (VGP cut-off angle 40°), which is in accord with the predicted value of approximately 12.5° for this latitude (McElhinny and McFadden, 1997). The VGP dispersion of the flows in this study is largely consistent with both anisotropic paleosecular variation models of Constable and Johnson (1999), because their axial anisotropy model and general anisotropy model yield approximately similar VGP dispersions at this location. It should be kept in mind that the profiles do not represent a continuous section in time, and that only ∼ 3 samples were collected from each flow unit in this reconnaissance study. Furthermore, several flow units are hampered by large uncertainties as the within-flow scatter is considerable, which can explain the large ASD value of 14.4°(+7.4/ − 3.7) in the Ribeira da Barca profile. Low ASD values of 8.3 °(+3.2/ − 1.8) and 8.6 °(+4.9/ − 2.3) for the São Gonçalo and Porto Formosa profile, respectively, indicate that the time spans covered by these profiles may be too short to correctly estimate the secular variation. Moreover, it seems possible that the São Gonçalo and Porto Formosa profiles may represent flow fields that were rapidly emplaced, which, as indicated by several earlier studies (e.g. Riisager et al., 2003), could further underestimate the PSV.

Only the paleomagnetic pole for the flows of reverse polarity in the São Gonçalo profile is indistinguishable from the geographic pole (Fig. 5d). The poles generally cluster around a mean direction some ∼ 9° off the geographic pole (7° if the Ribeira da Barca flows are excluded), indicating the presence of a systematic far-sidedness. The mean inclination of all non-transitional flows is 18.9 ° (24.9° if the Ribeira da Barca flows are excluded) (Table 3), which is 9.3° lower than the GAD value for this area. Such indications of a persistent quadrupole field must be considered highly tentative, however, because the paleomagnetic poles are not sufficiently well constrained. The main problem arises from the low number of flow units displaying non-transitional polarity, which implies that field variations may not be adequately represented to average out the secular variation. This problem may be more pronounced if the profiles actually represent lava flow fields that were rapidly emplaced. The fact that the mean directions obtained from several flow units are poorly defined also casts doubt on the reliability of the observed far-sidedness of the mean paleomagnetic pole.

5.2. Identification of cryptochron C2r.2r-1

The São Gonçalo profile contains a sequence of flows (SG-I) that display transitional polarity (Fig. 7). The uppermost and lowermost flows in event SG-I are temporally constrained to 2.41 ± 0.04 Ma and 2.38 ± 0.05 Ma, respectively. This implies that event SG-I occurred in the interval 2.37–2.43 Ma, or, since the ages are indistinguishable from one another at the 95% confidence level, that the sequence of five flows erupted over a period of time

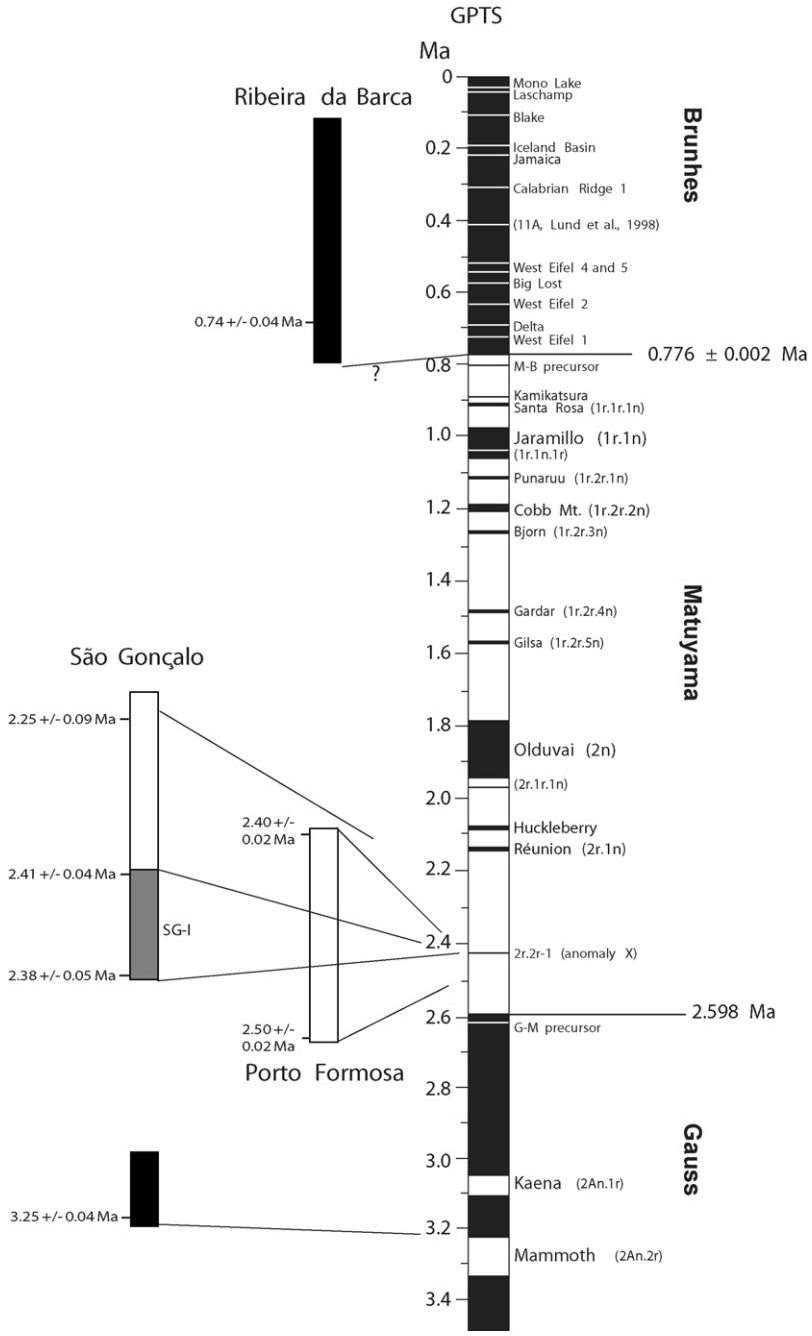


Fig. 8. Correlation between the São Gonçalo, Ribeira da Barca, Porto Formosa profiles and the Geomagnetic Polarity Time Scale. Correlation of the individual profiles are constrained to the GPTS by $^{40}Ar/^{39}Ar$ ages (Holm et al., 2008). Black corresponds to normal polarity, white to reverse polarity. Included in the GPTS are subchrons, cryptochrons, and geomagnetic excursions that are temporally well constrained and well-documented in the literature.

shorter than our best analytical precision, giving event SG-I an age of 2.40 ± 0.03 Ma. Considering that the 15 directions obtained from the five flows that define SG-I cluster around a common mean direction, it seems likely that the five flows erupted over a short period of time, possibly during a single eruptive event. The mean age and mean direction may consequently provide the most accurate representation of event SG-I. The timing of event SG-I indicates that it recorded cryptochron C2r.2r-1 (2.420–2.441 Ma) in the lower Matuyama Chron (Cande and Kent, 1995), since the timing does not match any other well-documented and temporally well-constrained cryptochrons or excursions. Given their synchronicity, we suggest that event SG-I corresponds to cryptochron C2r.2r-1 (Fig. 8).

Cryptochron C2r.2r-1 was first observed as a tiny wiggle in marine magnetic anomaly profiles with high spreading rates in the South Pacific (Heirtzler et al., 1968). Heirtzler et al. (1968) referred to it as "anomaly X", and an average age of 2.3 ± 0.1 Ma was subsequently assigned to this event (Emilia and Heinrichs, 1969). Using the terminology associated with the Geomagnetic Polarity Time Scale, Cande and Kent (1992) referred to anomaly X as cryptochron C2r.2r-1, and they constrained the anomaly to the interval 2.420–2.441 Ma (Cande and Kent, 1992, 1995).

Currently, there is only weak magnetostratigraphic evidence for the existence of the elusive anomaly X, cryptochron C2r.2r-1. A study of Chinese loess-paleosol sediments found excursions at 2.35 Ma and 2.42 Ma, the latter of which the authors interpret as

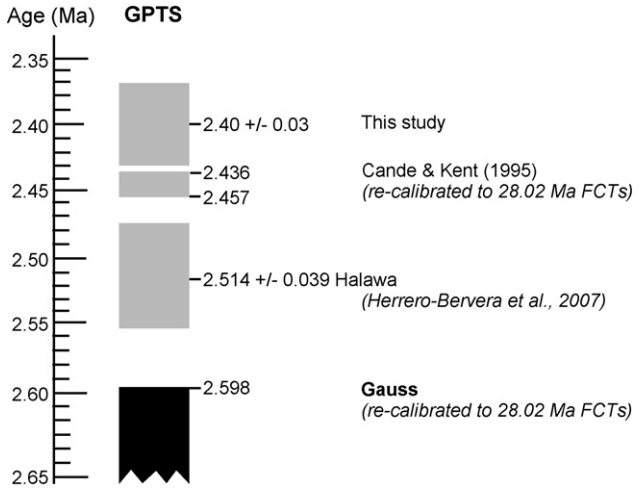


Fig. 9. A zoom of the GPTS around the interval encompassing cryptochron C2r.2r-1. The figure shows a comparison in time between the anomaly originally reported by Cande and Kent (1995), and the two well-dated anomalies recorded by lava flows that are likely to represent cryptochron C2r.2r-1.

cryptochron C2r.2r-1 (Yang et al., 2007). The estimated age was obtained by assuming a constant accumulation rate between polarity boundaries. A recent study of a lava sequence at Halawa, Oahu, Hawaii, identified an excursion, characterized by four flow units displaying transitional polarity, which apparently corresponds to cryptochron C2r.2r-1 (Herrero-Bervera et al., 2007). The age of the excursion observed in the Halawa record was determined by nine $^{40}Ar/^{39}Ar$ ages spanning the entire lava sequence (28 flow units), which proved indistinguishable from one another. This gives the whole section a weighted mean age of 2.514 ± 0.039 Ma, whereas the weighted mean age of the flows that recorded the event itself is somewhat higher (2.58 ± 0.12 Ma). Using an age of 28.02 Ma for the Fish Canyon Tuff (FCT) sanidine (Renne et al., 1998), Herrero-Bervera et al. (2007) re-calibrated the original age estimate of cryptochron C2r.2r-1 from Cande and Kent (1995), obtaining a new age interval of 2.436–2.457 Ma. The re-calibrated age of cryptochron C2r.2r-1 is in good agreement with the age of the transitional directions recorded in this study, which are based on the calibration value for the FCT sanidine used by Renne et al. (1994). The small difference between the calibration value for the FCT sanidine used in this study and the study of Herrero-Bervera et al. (2007) is insignificant. We note that there is no age overlap between the directional anomaly from this study (SG-I) and the slightly older Halawa excursion. (Fig. 9). This suggests that either cryptochron C2r.2r-1 was time-transgressive and that both SG-I and the Halawa excursion recorded this cryptochron or, alternatively, that SG-I and the Halawa excursion recorded two separate events.

Event SG-I from the São Gonçalo profile does not provide a full history of field changes during cryptochron C2r.2r-1. It is consequently impossible from this study alone to assess whether the field briefly reached a full normal-polarity state during cryptochron C2r.2r-1, but the Halawa record from Hawaii indicate that this was not the case. The $^{40}Ar/^{39}Ar$ ages from this do not improve the estimate of the duration of this cryptochron (21 kyr) from Cande and Kent (1995), and they provide no constraint on its minimum duration. However, if event SG-I and the Halawa sequence recorded the same anomaly, then the combined $^{40}Ar/^{39}Ar$ ages indicate a minimum duration of 45 kyr (2.430–2.475 Ma) for cryptochron C2r.2r-1.

5.2.1. Geomagnetic field behavior during cryptochron C2r.2r-1

Cryptochrons can be modelled either as very short polarity intervals or as slightly longer intensity fluctuations of the dipole field (Cande and Kent, 1992). The results presented here, combined with those obtained from the Halawa sequence (Herrero-Bervera et al., 2007) which possibly recorded the same anomaly, strongly suggest that cryptochron C2r.2r-1 was a global event associated with directional changes of the geomagnetic field. The VGPs associated with event SG-I in the São Gonçalo profile plot around Australia (Fig. 7), and hence very close to the paleomagnetic transitional VGP-cluster patches of Hoffman (1992). The VGPs are plotted using the paleomagnetic convention (Prevot and Camps, 1993). Hoffman (1992) argued that the transitional field displays rapid stop-and-go movements between meta-stable transitional-field states that are related to long-standing thermodynamic features of the core–mantle system. According to Hoffman (1992), the transitional field will spend most of its time within one of these hang-up points, which coincide with regions of fast seismic wave propagation in the lower mantle (Dziewonski and Woodhouse, 1987; Becker and Boschi, 2002). The results presented here suggest that the geomagnetic field may have been in a meta-stable transitional state during cryptochron C2r.2r-1, and that it spent some time in, or close to, one of these cluster patches around Australia. We note that the VGPs associated with the Halawa excursion, which cluster in the West Indian Ocean off the coast of Madagascar (Fig. 7), do not coincide with the VGP cluster associated with SG-I or with regions characterized by fast seismic wave propagation in the lower mantle. This implies that either the field morphology during cryptochron C2r.2r-1 was more complex than a simple dipole-dominated field structure, or that event SG-I and the Halawa excursion recorded two separate events.

6. Conclusions

New paleomagnetic results from three lava sequences on Santiago, Cape Verde Islands, have been presented. The profiles are constrained to the GPTS by $^{40}Ar/^{39}Ar$ ages. The Ribeira da Barca profile consists of Brunhes-aged lava flows of normal polarity and it represents the youngest profile in this paleomagnetic study. The Porto Formosa profile is constrained to the lower part of the Matuyama Chron by two $^{40}Ar/^{39}Ar$ ages, which is consistent with all flows in the profile displaying reverse polarity. The São Gonçalo profile consists of two sections. The uppermost section, which is dominated by reverse polarity flows, is constrained to the lower Matuyama and quite possibly overlap with the Porto Formosa profile. The lowermost section, consisting of normal polarity flows, is constrained to an interval, between the Mammoth and Kaena Subchrons, in the Gauss Chron.

The VGP dispersion of all non-excursional flows in the three profiles is $11.7\,^{\circ}(+2.7/-1.9)$, consistent with the predicted value for this latitude (McElhinny and McFadden, 1997) and both anisotropic paleosecular-variation models of Constable and Johnson (1999).

One event (SG-I), recording transitional polarity, was observed in the São Gonçalo profile. Event SG-I (5 flows) is constrained to the interval 2.37–2.43 Ma ($\pm 2\sigma$), which is in accord with the timing of cryptochron C2r.2r-1 in the marine magnetic anomaly record. Hence, we propose that event SG-I represent cryptochron C2r.2r-1, and that the cryptochron was a global-scale phenomenon. The VGPs associated with event SG-I suggest that the geomagnetic field may have been in a meta-stable transitional state during cryptochron C2r.2r-1 and spent some time in or around the paleomagnetic transitional VGP clusters around Australia.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.pepi.2009.01.009.

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